PN7150 NFC

U1
PN7150B0HN_C11002E-PN7150B0HN_C11002E

+3V3
R1 10K
R2 10K
SDA
SCL

RXP 16 — RXP
RXN 15 — RXN
ADR0 1 — I2CADR0
ADR1 3 — I2CADR1
SCL 7 — I2CSCL
SDA 5 — I2CSDA

27.12Mhz
Y1
C3 10pF
C4 10pF
VEN 10 — VEN
NFC_CLK_XTAL1 36
NFC_CLK_XTAL2 37

VDDD 30
VDD 29
VDDA 28
VDD(TX) 14
VBAT 26
VBAT1 13
VBAT2 12
VDD(TX_IN) 22
VDD(MID) 17
VDD(PAD) 6
C5 1uF
C6 1uF
C7 100nF
C8 1uF
C9 100nF
TX1 21 — TX1
TX2 18 — TX2
CLK_REQ 40 — CLK
IRQ 8 — IRQ
VSS(TX) 19
VSS(PAD) 4
VSS
GND

LEDS & BUTTONS

SW1 RESET — RST — GND
GPIO SW2 — GPIO1 — GND
R13 1K — D3 LED — LED1 — GND
R12 1K — D2 LED — ESP_LED — GND

ANTENA MACHING CIRCUIT

RXP — C23 1nF/16V — L1 160nH — R15 2k2 — TX1
TX1 — 100pF/50V C27
C25 680pF — C30 120pF — R18 3.24R — L3 L
C26 680pF — C31 120pF — R19 3.24R
TX2 — L2 160nH — R16 2k2 — 100pF/50V C28 — RXN
RXN — C24 1nF/16V
GND

FLASH

+3V3
10K R20
QSPI_CS
QSPI_SCK
C33 100nF
GND

U6 W25Q16JV
CS 1
CLK 6
DI(IO0) 5 — QSPI_DATA0
DO(IO1) 2 — QSPI_DATA1
IO2 3 — QSPI_DATA2
IO3 7 — QSPI_DATA3
VCC 8
GND 4
EP 9

MICROCONTROLLER

U2
+1V2
+3V3
DVDD 50
DVDD 23
VREG_VOUT 45
VREG_IN 44
IOVDD 49
IOVDD 42
IOVDD 33
IOVDD 22
IOVDD 10
IOVDD 1
USB_VDD 48
ADC_AVDD 43
USB_DP 47 — R9 27 — D+
USB_DM 46 — R10 27 — D−
RD+
RD−

QSPI_CS 56 — QSPI_SS
QSPI_DATA0 53 — QSPI_SD0
QSPI_DATA1 55 — QSPI_SD1
QSPI_DATA2 54 — QSPI_SD2
QSPI_DATA3 51 — QSPI_SD3
QSPI_SCK 52 — QSPI_SCLK

Raspberry Pi
RP2040

XIN 20
Y2 12Mhz
C1 27pF
C2 27pF
XOUT 21 — 1K R3
RUN 26 — RST
SWCLK 24 — SWCLK
SWDIO 25 — SWD
TESTEN 19
GND 57

GPIO0 2
GPIO1 3
GPIO2 4
GPIO3 5 — ESP_RESET
GPIO4 6
GPIO5 7
GPIO6 8 — VEN
GPIO7 9 — IRQ
GPIO8 11 — CIPO_TX
GPIO9 12 — CS_RX
GPIO10 13 — ACK_CTS
GPIO11 14 — COPI_RTS
GPIO12 15 — SDA
GPIO13 16 — SCL
GPIO14 17 — SCK
GPIO15 18
GPIO16 27
GPIO17 28 — GPIO1
GPIO18 29 — IN_B
GPIO19 30 — IN_A
GPIO20 31 — ESP_GPIO0
GPIO21 32
GPIO22 34
GPIO23 35
GPIO24 36
GPIO25 37
GPIO26_ADC0 38 — B_ADC
GPIO27_ADC1 39 — LED1
GPIO28_ADC2 40
GPIO29_ADC3 41

JP1 BOOT — QSPI_CS — GND

+3V3: C10 1uF, C12 100nF, C13 100nF, C15 100nF, C17 100nF, C19 100nF, C20 100nF, C21 100nF
+1V2: C14 1uF, C16 100nF, C18 100nF
GND

WIFI & BLE NINA

+3V3
R17 10K
ESP_RESET

U5 ESP32-WROOM-32D
EN 3
VDD 2
SENSOR_VP 4
SENSOR_VN 5
SDO/SD0 21
SDI/SD1 22
SHD/SD2 17
SWP/SD3 18
SCK/CLK 20
SCS/CMD 19
GND 1

IO0 25 — ESP_GPIO0
TXD0/IO1 35 — CS_RX
IO2 24
RXD0/IO3 34 — CIPO_TX
IO4 26
IO5 29 — CS_RX
IO12 14 — COPI_RTS
IO13 16
IO14 13
IO15 23
IO16 27
IO17 28
IO18 30 — SCK
IO19 31
IO21 33
IO22 36
IO23 37 — CIPO_TX
IO25 10 — ESP_LED
IO26 11
IO27 12
IO32 8
IO33 9 — ACK_CTS
IO34 6
IO35 7

+3V3 — C32 100nF — GND

COIL

VBUS
+BATT
SW4A SOURCE MAG

U7 TC4424
NC 1
IN_A 2
GND 3
IN_B 4
NC 8
OUT_A 7 — OUTA
VDD 6
OUT_B 5 — OUTB

C34 10uf
GND

OUTA 1
OUTB 2
J3 MAGSPOOF

REGULATOR AND BATTERY

P1 USB_C_Plug_USB2.0
VBUS A4
CC A5 — R4 5k1
VCONN B5 — R5 5k1
D− A7 — D−
D+ A6 — D+
SHIELD
GND A1
S1
F1 0.5A — VBUS

+BATT — R6 100K — B_ADC
R7 100K — GND

VBUS — U3 MCP73831-3-OT — +BATT
R8 5.1K — PROG
VDD 4
VSS 2
VBAT 3 — C11 10uF
STAT 1 — R11 1K GND — D1 LED — VBUS
GND

+BATT — J1 Battery — GND

VBUS — D4 MBR120 — R14 100K — GND
+BATT — Q1 DMG3415U-7
ON — SW3
C22 10uF — GND

U4 AP2112K-3.3
VIN 3
EN 1
VOUT 5
GND 2
+3V3 — C29 10uF — GND

ADDRESS SELECT

R21 100K — ADR0 — GND
R22 100K — ADR1 — GND

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

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